

A MINI PROJECT ON

Finance Management Model Bot

Submitted for the partial fulfillment of the requirement for the degree of

Bachelors of Business Administration in Artificial Intelligence with Honors

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(Semester – 1)

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CERTIFICATE

This is to certify that the mini project titled **Finance Management Model Bot** has been carried out by **Ashwin Chhawaniya (25030422028)**, **Arnav Singh (25030422022)**, and **Akshat Goyal (25030422014)** of **Bachelor of Business Administration in Artificial Intelligence with Honors (Semester – 1)**, in partial fulfilment of the requirements for the award of the degree of **Bachelor of Business Administration in Artificial Intelligence with Honors**.

The project has been completed under the supervision and guidance of **Dr. Dawa Chyophel Lepcha, Assistant Professor – Python Programming**, at the **Symbiosis Artificial Intelligence Institute, a Constituent of Symbiosis International (Deemed University), Lavale, Mulshi, Pune – 411211, Maharashtra**, during the academic year **2025**.

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DECLARATION

We hereby declare that the mini project titled **FINANCE MANAGEMENT MODEL BOT** has been carried out by **ASHWIN CHHAWANIYA (25030422028)**, **ARNAV SINGH (25030422022)**, and **AKSHAT GOYAL (25030422014)** of **BACHELOR OF BUSINESS ADMINISTRATION IN ARTIFICIAL INTELLIGENCE WITH HONORS (SEMESTER – 1)**, under the guidance of **DR. DAWA CHYOPHEL LEPCHA, ASSISTANT PROFESSOR – PYTHON PROGRAMMING**, at the **SYMBIOSIS ARTIFICIAL INTELLIGENCE INSTITUTE (SAII)**, a constituent of **SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)**, Lavale, Mulshi, Pune – 411211, Maharashtra.

We further declare that this project report is the result of our original work and has not been submitted to any other university or institution for the award of any degree or diploma. All sources of information used in the preparation of this report have been duly acknowledged.

Place: SYMBIOSIS ARTIFICIAL INTELLIGENCE INSTITUTE (SAII), SIT BUILDING, 6TH FLOOR

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ABSTRACT

The mini project titled **FINANCE MANAGEMENT MODEL BOT** is developed using **PYTHON** as the core programming language and **MYSQL** as the database for data storage and management. The primary objective of this project is to design an intelligent, user-friendly financial assistant that helps users efficiently manage, analyze, and forecast their finances.

The system allows users to record income and expenses, plan monthly budgets, calculate taxes, compute compound interest, and perform real-time currency conversions. It also provides visual insights through graphical representations using **MATPLOTLIB**, thereby enhancing decision-making based on financial trends and savings growth.

The project demonstrates the effective integration of **PYTHON PROGRAMMING**, **DATABASE CONNECTIVITY**, and **DATA VISUALIZATION** techniques to create a functional, scalable, and interactive finance management tool. It serves as a practical implementation of theoretical knowledge and highlights the growing role of automation and artificial intelligence in the field of financial management.

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CHAPTER 1 – INTRODUCTION

1.1 Background:

The **Finance Management Model Bot** is a Python-based command-line application created to assist students in managing financial activities such as income tracking, expense logging, tax computation, and savings analysis.

It demonstrates a practical application of Python programming concepts, MySQL database connectivity, and data visualization tools.

1.2 Problem Definition:

Manual financial tracking methods are error-prone and time-consuming. This system provides a reliable, automated, and user-friendly tool that helps students plan budgets, calculate savings, and visualize long-term financial outcomes.

1.3 Scope of the Project:

The bot enables students to:

- Track and categorize financial transactions
- Set and manage budgets
- Compute tax and investments
- Convert currencies using live rates
- Save and analyze data persistently via MySQL

CHAPTER 2 – LITERATURE REVIEW / BACKGROUND

Financial management applications are widely used in modern computing. However, existing solutions are often complex or paid, limiting accessibility for students.

This project combines core financial calculations (budgeting, taxation, investment) with database and visualization tools into a single Python program. The objective is educational: to demonstrate how automation can enhance real-world financial literacy and data handling skills.

CHAPTER 3 – SYSTEM ANALYSIS AND DESIGN

3.1 Hardware and Software Requirements:

Hardware:

- Processor: Intel Core i3 or higher
- RAM: Minimum 4 GB
- Storage: 500 MB free space

Software:

- Python 3.7 or higher
 - MySQL Server 5.7 or higher
 - Required Packages: `mysql-connector-python`, `matplotlib`, `forex-python`
-

3.2 Database Design:

The system uses **college_finance_db**, which includes:

- `users`: Login details
- `entries`: Income/expense records
- `monthly_budget`: Budget allocations
- `bill_splits`: Shared expenses
- `tax_calculations`: Tax results
- `compound_interest_records`: Investment calculations

- `currency_conversions`: Exchange history
- `savings_records`: Savings projections

Each table is linked to `user_id` for secure and personalized data storage.

CHAPTER 4 – IMPLEMENTATION AND WORKING

4.1 Implementation Tools:

- **Programming Language:** Python 3.x
 - **Database:** MySQL
 - **Visualization:** matplotlib
 - **API Integration:** forex-python
 - **Database Connector:** mysql-connector-python
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4.2 Functional Modules:

- **User Management:** Handles registration and login.
 - **Budget Planner:** Allocates monthly income percentages.
 - **Expense Tracker:** Logs categorized expenses.
 - **Bill Splitter:** Divides bills equally among users.
 - **Tax Calculator:** Computes taxes as per Indian tax slabs.
 - **Compound Interest Calculator:** Projects investment growth.
 - **Savings Calculator:** Estimates total balance over time.
 - **Currency Converter:** Performs real-time and offline conversions.
-

CHAPTER 5 – RESULTS AND DISCUSSION

All modules were tested for accuracy and stability.

Testing confirmed:

- Correct database record creation.
- Accurate tax and savings calculations.
- Functional live and offline currency conversion.
- Proper graphical output via matplotlib.

Performance was smooth with negligible delay in database queries. The system is scalable for further enhancements.

CHAPTER 6 – CONCLUSION AND FUTURE SCOPE

The **Finance Management Model Bot** successfully integrates key financial management functionalities into a single Python-based application.

Future enhancements:

- Web and mobile interface
 - Banking API integration
 - Automated expense categorization using AI
 - Password encryption and multi-user dashboard
-

REFERENCES

1. Python Software Foundation – <https://docs.python.org/>
2. MySQL Documentation – <https://dev.mysql.com/doc/>
3. matplotlib Library – <https://matplotlib.org/>
4. forex-python Package – <https://pypi.org/project/forex-python/>

5. GitHub Project Repository – <https://github.com/java-mx/college-finance-bot>

APPENDIX – SUMMARY OF SOURCE CODE

Below are representative excerpts from the main Python source file (`Finance_Bot_FMM.py`), showing essential modules and logic structure.

1. Database Connection and Setup:

```
import mysql.connector

DB_CONFIG = {
    "host": "localhost",
    "user": "root",
    "password": "your_own_password",
    "database": "college_finance_db"
}

def db_connect():
    cnx = mysql.connector.connect(**DB_CONFIG)
    cursor = cnx.cursor()
    return cnx, cursor
```

2. User Registration and Login:

```
def register_user():
    cnx, cursor = db_connect()
    username = input("Choose a username: ").strip()
    password = input("Choose a password: ")
    cursor.execute(
```

```

        "INSERT INTO users (username, password_hash) VALUES (%s, %s)",
        (username, password)
    )
    cnx.commit()
    print("Registration successful!")
    cursor.close()
    cnx.close()

def login_user():
    cnx, cursor = db_connect()
    username = input("Enter username: ").strip()
    password = input("Enter password: ")
    cursor.execute("SELECT id, password_hash FROM users WHERE username = %s",
    (username,))
    row = cursor.fetchone()
    if row and password == row[1]:
        print(f"Welcome back, {username}!")
        return row[0]
    else:
        print("Invalid credentials.")
        return None

```

3. Monthly Budget Planner:

```

def monthly_budget_planner(user_id):
    cnx, cursor = db_connect()
    budget = float(input("Enter total monthly budget: "))
    priorities = [input(f"Enter priority #{i+1}: ") for i in range(3)]
    percents = [int(input(f"Enter % for {p}: ")) for p in priorities]
    misc_pct = 100 - sum(percents)
    cursor.execute(
        "INSERT INTO monthly_budget (user_id, month, budget, priority1,
        priority1_percent, priority2, priority2_percent, priority3,
        priority3_percent, miscellaneous_percent) VALUES
        (%s,%s,%s,%s,%s,%s,%s,%s,%s,%s)",
        (user_id, "2025-11", budget, priorities[0], percents[0],
        priorities[1], percents[1], priorities[2], percents[2], misc_pct)
    )
    cnx.commit()
    cursor.close()

```

```
cnx.close()
```

4. Tax Calculator:

```
def tax_calculator(user_id):
    cnx, cursor = db_connect()
    income = float(input("Enter annual income (₹): "))
    slabs = [(250000, 0), (250000, 0.05), (500000, 0.20), (float("inf"),
0.30)]
    remaining = income
    tax = 0
    for slab, rate in slabs:
        taxable = min(remaining, slab)
        tax += taxable * rate
        remaining -= taxable
        if remaining <= 0:
            break
    effective_rate = (tax / income) * 100
    print(f"Tax due: ₹{tax:.2f}")
    print(f"Effective rate: {effective_rate:.2f}%")
    cursor.execute(
        "INSERT INTO tax_calculations (user_id, income, tax, effective_rate)
VALUES (%s,%s,%s,%s)",
        (user_id, income, tax, effective_rate)
    )
    cnx.commit()
```

5. Currency Converter:

```
from forex_python.converter import CurrencyRates
```

```
def currency_converter(user_id):
    c = CurrencyRates()
```

```
amount = float(input("Enter amount: "))
from_curr = input("From currency code: ").upper()
to_curr = input("To currency code: ").upper()
converted = c.convert(from_curr, to_curr, amount)
print(f"{amount:.2f} {from_curr} = {converted:.2f} {to_curr}")
```

6. Main Function:

```
def main():
    create_all_tables()
    print("Welcome to Finance Management Model Bot")
    user_id = None
    while True:
        choice = input("Are you a new user or returning? (new/login): ").lower()
        if choice == "new":
            register_user()
        elif choice == "login":
            user_id = login_user()
            if user_id:
                break
        while True:
            print("1. Budget Planner\n2. Bill Splitter\n3. Tax Calculator\n4. Compound Interest\n5. Currency Converter\n6. Savings\n7. Record Income/Expense\n8. Logout")
            option = input("Select: ")
            if option == "1": monthly_budget_planner(user_id)
            elif option == "3": tax_calculator(user_id)
            elif option == "5": currency_converter(user_id)
            elif option == "8": break
```

Note:

The full source code (Finance_Bot_FMM.py) and SQL database schema (College_Finance_DB.sql) are attached separately as supplementary files.
